

PORTABLE ANTENNA ROTATOR (PAR)

Interface Control Document

Version 1.0

04/29/2026

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1. Introduction

This ICD describes the relationship between:

- PAR electrical box and the PAR rotator structure.
- PAR electrical box and the external power source.
- Raspberry Pi 4B (RPi) and the connected sensors, motors, touchscreen, and relays.
- PAR and the operator.

This ICD specifies the interface requirements which the participating systems must meet. It describes the concept of operations for the interface, defines the protocols that govern the interchange of data, and identifies the communication paths along which the project team expects signals to flow.

For each interface, the ICD provides the following information:

- A description of the data exchange format and protocol for exchange
- A general description of the interface
- Assumptions where appropriate

1.1 Overview

The interface between the electrical box and the rotator structure exists to convey power and control signals between the rotator structure and the electrical box. A 10 ft DSUB cable provides power for the motors, control signals for the motors, I2C signals for the magnetometer / accelerometer, and analog voltages from the potentiometer on the rotator structure.

The interface between the electrical box and the external power source exists to provide a way to power PAR and distribute power to additional equipment. An Anderson power pole bus connected to the outer wall of the electrical box provides a way for the operator to connect a battery. Additional equipment may also be connected to this bus to receive power from the battery.

The interface between the RPi and the connected sensors, motors, touchscreen, and relays exists to provide the RPi with data about the condition of the electrical box and its surroundings, steer the rotator structure in a desired direction, and control indication lights. Dupont connectors link RPi GPIO pins to voltage buses and pin connections on the sensors.

The interface between PAR and the operator exists to provide a way for the operator to input data, see RPi operation, manage files, analyze satellite passes, and control the rotator structure.

2. Assumptions/Constraints/Risks

It is assumed that the operator will provide their own power through a power pole connection.

2.2 Constraints

Connection to the rotator structure must be done with the provided 17 pin DSUB cable. This cable must only be connected to the PAR electrical box to avoid damaging components as the motors will draw significant current.

The power source connected to the power pole bus must have a voltage within 9 – 32 VDC. Whenever voltage from the source degrades below 9 VDC, the system will lose power. The operator should take this into consideration when connecting a battery.

3. Interface Overview

3.1 Interface Overview

The electrical box-rotator structure interface conveys power and control signals between the electrical box and the rotator structure. The interface is made through a 17-pin, 10 ft DSUB cable. This cable provides 12 VDC power for the elevation servo motor, 5 VDC for the azimuth control feedback potentiometer, and 3 VDC for an accelerometer. Alongside power distribution, the DSUB cable also relays I2C data between the accelerometer and the RPi, analog data from the potentiometer to an analog to digital converter in the electrical box, PWM signals from the RPi to the servo motor, and pulse control signals to the azimuth stepper motor from the stepper motor driver.

The interface between the electrical box and the external power source provides a way to power the Portable Antenna Rotator (PAR) and distribute power to additional equipment. The interface is made through a six-pole, twelve terminal power pole bus that is connected to the outer wall of the electrical box. This bus allows the operator to connect a battery through the power pole connectors. Allowable voltages on this bus are 9 VDC to 32 VDC. Power from the power pole connection is distributed to both the control / operator interface circuitry inside the electrical box as well as to the motors and electrical components on the rotator structure. Additional equipment separate from PAR can be powered from the power pole bus by connecting the equipment in parallel to the battery.

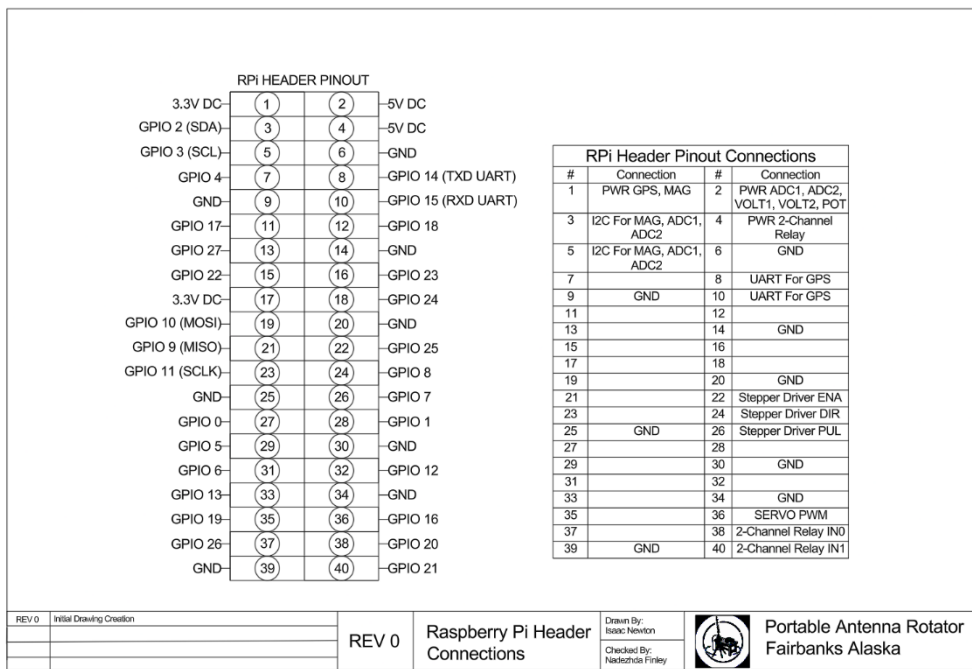


Figure 1: PAR-004 Design Drawing, RPi Header Diagram

The interface between the RPi and the sensors, motors, touchscreen, and relays provides the RPi a way to perceive information about PAR and its surroundings, control the rotator structure, and communicate system information to the operator. The RPi DSI FPC display port is used to connect the RPi to the touchscreen display, and the RPi 40-pin GPIO is used to connect the RPi to the various peripherals. Figure 1 above shows connections on the RPi GPIO for reference. Signal interfacing protocols on the GPIO include I2C, UART, PWM, and pulse signals written into the Portable Antenna Rotator Router (PARR) program. +3 VDC and +5 VDC power for low current draw sensors are also sourced from the RPi GPIO. Devices connected to the RPi include: two USB extension cables, a 7 inch touchscreen display, two analog to digital converters, two voltage divider “voltmeters”, a GPS, an accelerometer / magnetometer, a servo motor, a stepper motor driver, and a relay for the LED indication lights. Dupont connectors link RPi GPIO pins to voltage buses and pin connections on the sensors.

The interface between PAR and the operator provides a way for the operator to input data, visualize RPi operation, manage files, interact with satellite tracking software, and control the rotator structure. The interface is accomplished through a touch screen, two accessible USB-A ports, a power switch, and an emergency power-off button. The touchscreen provides a way for the operator to interact with the PARR program, communicate with the controller, and steer the rotator structure. The USB-A ports allow the operator to connect a USB storage device, mouse, or keyboard to the RPi for file upload and interfacing. The power switch and emergency power-off button allow the operator to control the flow of power to PAR. The switch connects and disconnects power from the power pole bus to PAR, and the emergency power-off button is a resettable button that disconnects power from the RPi and motors and turns on a green LED light while compressed.

3.1.1 Interface Controls

Table 1: Electrical Box to Rotator Structure Interface Connections

Interface Type	Interface From	Interface To	Description of Interface	Other Information
DSUB Connector	PAR rotator structure	PAR electrical box	Connects 12 VDC power, 5 VDC power, 3 VDC power, stepper motor control signals, PWM signals, I2C signals, and analog potentiometer voltage between the rotator structure and RPi	

Table 2: Electrical Box to External Power Source Interface Connections

Interface Type	Interface From	Interface To	Description of Interface	Other Information
9 - 32 VDC power	Power source	Electrical box	Anderson Power Pole Connectors	

Table 3: RPi to Sensors, Motors, Touchscreen, and Relay Interface Connections

Interface Type	Interface From	Interface To	Description of Interface	Other Information
RPi GPIO	Board Pin # 1	Magnetometer, GPS	3.3 VDC Power	
RPi GPIO	Board Pin # 2	ADC1, ADC2, Voltmeter1, Voltmeter2, Potentiometer	5 VDC Power	

RPi GPIO	Board Pin # 3	Magnetometer / Accelerometer, ADC1, ADC2	SDA	
RPi GPIO	Board Pin # 4	2CH Relay (LED)	5 VDC Power	
RPi GPIO	Board Pin # 5	Magnetometer / Accelerometer, ADC1, ADC2	SCL	
RPi GPIO	Board Pin # 6	GND	GND	
RPi GPIO	Board Pin # 9	GND	GND	
RPi GPIO	Board Pin # 8	GPS RX	UART (RPi TX)	
RPi GPIO	Board Pin # 10	GPS TX	UART (RPi RX)	
RPi GPIO	Board Pin # 14	GND	GND	
RPi GPIO	Board Pin # 20	GND	GND	
RPi GPIO	Board Pin # 22	Stepper Motor Driver	Stepper Enable	
RPi GPIO	Board Pin # 24	Stepper Motor Driver	Stepper Direction	
RPi GPIO	Board Pin # 25	GND	GND	
RPi GPIO	Board Pin # 26	Stepper Motor Driver	Stepper Pulse	
RPi GPIO	Board Pin # 30	GND	GND	
RPi GPIO	Board Pin # 34	GND	GND	
RPi GPIO	Board Pin # 36	Servo Motor	PWM Signal	
RPi GPIO	Board Pin # 38	2CH Relay (LED)	Pulse Signal	
RPi GPIO	Board Pin # 39	GND	GND	
RPi GPIO	Board Pin # 40	2CH Relay (LED)	Pulse Signal	
RPi USB	RPi USB-A	USB Extension Cable		
RPi USB	RPi USB-A	USB Extension Cable		
RPi Display Port	RPi DSI Display Port	7 Inch Touchscreen Display	FPC Connector	

Figure 2 shows a KiCad schematic for the connections to the RPI GPIO.

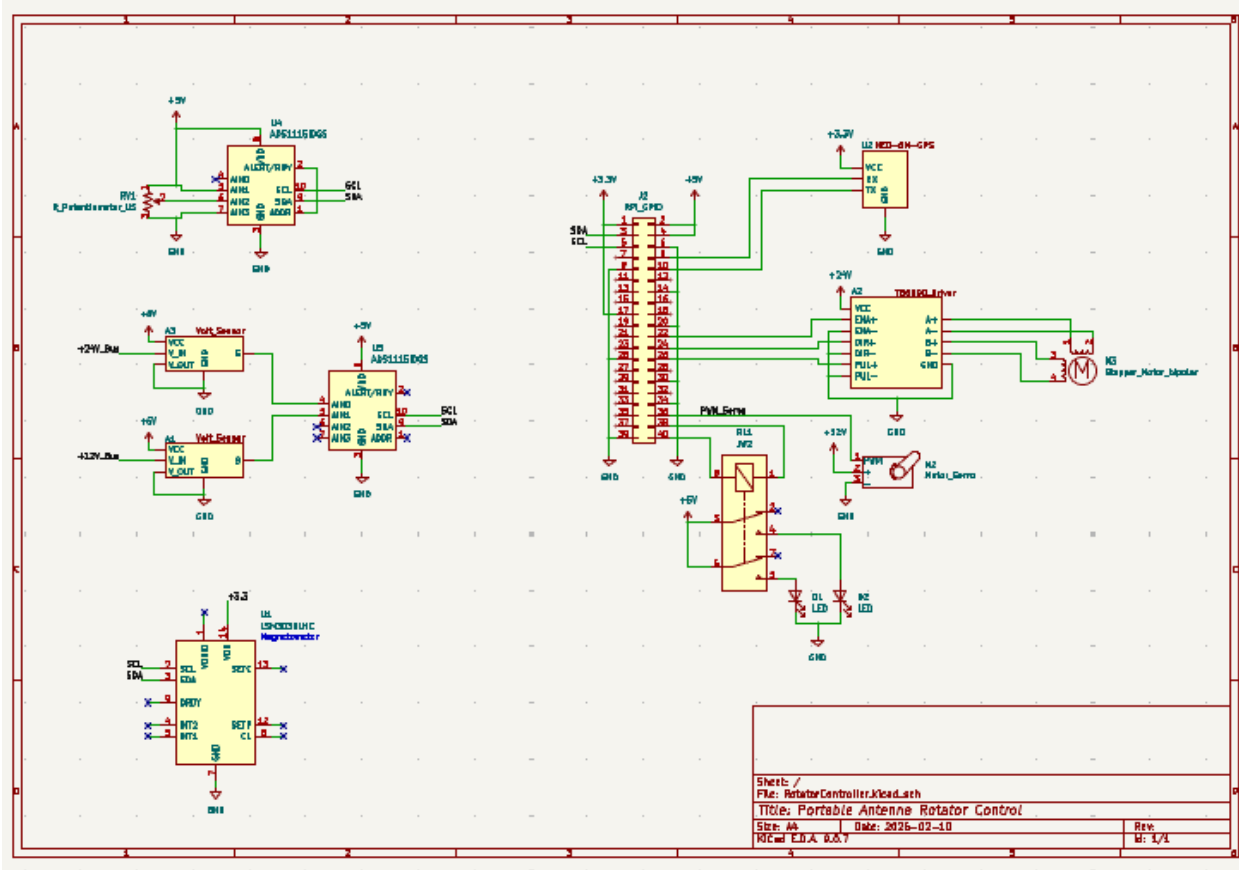


Figure 2: Portable Antenna Rotator RPi GPIO Connections, KiCad Schematic

Table 4: PAR Electrical Box to Operator Interface Connections

Interface Type	Interface From	Interface To	Description of Interface	Other Information
Touchscreen	Touchscreen	Operator	Provides the Operator with Visual Feedback and Tactile Control of the RPi	
USB Port 1	RPi USB Port	External USB-A Connector		
USB Port 2	RPi USB Port	External USB-A Connector		
Power Switch	Power Line	Operator	Allows Operator to Control Power Flow to PAR	
Emergency Power-off Button	Further Along the Power Line	Operator	Allows Operator to Remove and Restore Power to the RPi and Motors	

3.2 Functional Allocation

The function of the electrical box is to act as the control and power distribution center of PAR. The DSUB interface between the electrical box and the rotator structure supports the function of the electrical box by transmitting power and signals to components on the rotator structure. To engage the DSUB interface, the operator must uncoil the 10 ft cable from stowed position and plug the cable into the DSUB terminal on the electrical box.

The function of the electrical box to external power source interface is to provide a way for the operator to connect power to the PAR system and other devices connected to the power pole bus. To connect a power source to the power pole bus, the operator must connect cables terminated with Anderson power pole connectors to the power pole bus on the box. To power external devices off of the power pole bus, the operator must plug in devices to the bus using cables terminated with Anderson power pole connectors.

The function of the RPi to sensors and motors interface is to provide the RPi with a means of sensing its surroundings and to control the rotator structure. There is nothing that the operator must do to engage this interface for system operation.

The function of the PAR to operator interface is to provide a way for the operator to engage with the PAR system. To power on PAR, the operator will need to disengage the emergency power-off button and flip the power switch to the “ON” position. To navigate files, apps, and the PARR program, the operator must either utilize the touchscreen function of the display or connect a mouse and keyboard to the USB ports.

Appendix A: Record of Changes

This interface control document should be kept in a folder with other PAR related documentation to keep a record of interface connections and purposes for system maintenance and updates to system design.

Table 5: Record of Changes

Version Number	Date	Author	Description of Change
1.1	04/29/2026	Nadezhda Finley	First version; no changes
<X.X>	<MM/DD/YYYY>		<Description of Change>
<X.X>	<MM/DD/YYYY>		<Description of Change>
<X.X>	<MM/DD/YYYY>		<Description of Change>

Appendix B: Glossary

Table 6: Glossary

Term	Acronym	Definition
Portable Antenna Rotator	PAR	The portable antenna rotator is the antenna rotator system whose interfaces are described in the present document. PAR consists of an electrical box and a rotator structure.
Portable Antenna Rotator Router	PARR	Portable Antenna Rotator Router is the Python software program written to control PAR.